Restaurant Sales Analysis

Business Insights & Strategic Recommendations

Project Type: Business Intelligence Portfolio Project
Tools Used: SQL, Microsoft Excel, Power BI
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Executive Summary

This project presents an end-to-end Business Intelligence solution for analyzing restaurant sales using **MySQL, Microsoft Excel, and Power BI**. The objective was to transform raw transactional data into actionable insights that support operational efficiency and data-driven decision-making.

The analysis covered **267 restaurant transactions**, generating **$8,454.01** in total sales, with an average bill of **$31.66**, an average party size of **2.56**, and an average tip of **18.27%**. The project included data cleaning, exploratory data analysis (EDA), KPI development, business analysis, management reporting, and advanced SQL techniques.

Interactive dashboards and analytical reports were developed to evaluate sales performance, customer behavior, server productivity, payment preferences, and operational trends. The insights and recommendations produced through this analysis demonstrate how SQL, Excel, and Power BI can be integrated to support effective business intelligence and strategic decision-making.

Key Business Insights

1. **Sales Performance**: The restaurant generated **$8,454.01** from **267 transactions**, with an average bill of **$31.66**, demonstrating consistent customer spending.
2. **Peak Business Period**: Saturday generated the highest revenue, while Lunch was the best-performing meal period, indicating clear peak demand patterns.
3. **Customer Behavior**: Customers typically visited in small groups (average party size **2.56**) and maintained a healthy average tip of **18.27%**.
4. **Payment Preference**: Card payments represented the majority of sales (approximately **76%**), highlighting customers' preference for digital payment methods.
5. **Employee Performance**: Server performance varied considerably, with **Mac** contributing the highest revenue, demonstrating measurable differences in sales effectiveness.
6. **Business Intelligence Value**: Combining SQL, Excel, and Power BI transformed raw transactional data into actionable insights for operational and strategic decision-making.

Business Challenges

1. Revenue is heavily dependent on weekend performance, creating an uneven sales distribution.
2. Breakfast generates significantly lower revenue compared to Lunch and Dinner, indicating underutilized business hours.
3. Performance differences among servers suggest opportunities for coaching and standardization.
4. The current average bill presents opportunities to increase customer spending through upselling.
5. Continuous KPI monitoring is required to support proactive business decisions instead of reactive management.

Strategic Recommendations

1. Increase weekday sales through targeted promotions and customer loyalty campaigns.

2. Optimize staffing and inventory based on meal-time demand and daily sales trends.
3. Implement KPI-based performance reviews and coaching for restaurant staff.
4. Increase average order value through upselling, cross-selling, and meal bundle strategies.
5. Maintain interactive SQL, Excel, and Power BI dashboards for continuous business monitoring.
6. Use historical sales data to support forecasting, resource planning, and long-term strategic decision-making.

Skills Demonstrated

- SQL & Advanced SQL
- Data Cleaning & Transformation
- Exploratory Data Analysis (EDA)
- KPI Development
- Microsoft Excel
- Power BI Dashboard Development
- Data Visualization
- Business Analysis
- Management Reporting
- Business Intelligence

Conclusion

This project demonstrates an end-to-end Business Intelligence solution by integrating MySQL, Microsoft Excel, and Power BI to transform raw restaurant transaction data into actionable business insights. Through data cleaning, KPI development, business analysis, interactive dashboards, and management reporting, the project supports evidence-based decision-making and showcases the practical skills required for Business Analyst and Data Analyst roles.